

Inspiring Fashion through Data: Color Extraction from Ghibli Films

Report

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Table of Contents

Table of Contents.....	2
Introduction to the Project and Its Goals.....	3
Data Collection Process.....	4
Initial Attempt.....	4
Google Cloud Storage Solution.....	4
Optimized Approach.....	5
Data Processing and Analysis.....	6
Overall Film Analysis.....	6
Top 10 dominant colors across the films (graph).....	6
Segmented Analysis.....	7
Three Segments:.....	7
Top 10 dominant colors across the films in 3 segments (graph).....	7
Five Segments:.....	8
Top 10 dominant colors across the films in 5 segments (graph).....	8
Frame-by-Frame Analysis.....	10
Plotting each frames dominant color (graph).....	11
Segmented Frame Analysis.....	11
Howls - Plotting each frame's dominant color by segment (graph).....	12
Nausicaa - Plotting each frame's dominant color by segment (graph).....	14
K-Means Clustering.....	17
Elbow Method:.....	17
Cluster Analysis:.....	18
Clustering output at 5,10,15 clusters (graph).....	18
3D RGB Space Visualization:.....	19
Entire Film K-Means Clustering (graph).....	19
Segmented Film K-Means Clustering (graph).....	20
Refined Clustering:.....	22
Final Color Palette.....	23
Created Outfits.....	24
Howl's Created Outfits (Images).....	25
Nausicaa's Created Outfits (Images).....	28

Introduction to the Project and Its Goals

My project is a unique blend of my passion for fashion and love for Ghibli films, explored through the lens of data science. The goal of this project was to extract and analyze the color palettes from Ghibli films to inspire fashion outfits. By leveraging data science techniques, I aimed to identify the most prominent colors throughout the films and use these colors to create visually appealing and thematically consistent outfits. This project is an intersection of art, technology, and creativity, showcasing how data-driven approaches can be applied to the world of fashion design.

For the movies, I chose two of my favorite films: "Howl's Moving Castle" and "Nausicaä of the Valley of the Wind." Honestly, there weren't any specific reasons for selecting these films beyond the fact that they are some of my favorites and possess beautiful color palettes that I thought would be fun to work with.

This report aims to give an overview of what was done while presenting the graphs and final outputs rather than showing the code. The detailed code and explanations can be viewed in the Jupyter notebooks available on GitHub.

Data Collection Process

Initial Attempt

In my initial attempt to collect data, I used FFmpeg to extract 24 frames per second from the film. This process quickly consumed all available storage space on my PC, leading to a near-system crash. It became evident that storing every frame of the film was not a viable solution due to the significant storage requirements.

I initially thought that by capturing every frame, I could ensure no color information was lost. However, this approach was impractical due to the sheer volume of data generated. The storage requirements ballooned rapidly, highlighting the need for a more efficient solution.

Google Cloud Storage Solution

To address the storage issue, I decided to leverage Google Cloud Storage (GCS). The plan was to extract 100 frames at a time, upload them to GCS, and then delete them from my local storage. This batch-processing approach aimed to manage the data flow more effectively and prevent local storage from being overwhelmed.

Despite solving the local storage issue, the process was extremely slow. Each batch operation of extracting, uploading, and deleting frames was time-consuming, significantly delaying the data collection process. To mitigate this, I implemented multiprocessing, which allowed me to use multiple cores for simultaneous processing, uploading, and deleting of frames. This optimization enhanced the speed of data handling.

However, while this method improved the workflow, my brother, a data scientist, pointed out a critical flaw. He questioned the necessity of turning a 2 GB movie into 300 GB of data. This observation prompted me to reevaluate my approach. I realized that my strategy involved storing a lot of redundant information, which was inefficient and unnecessary.

Optimized Approach

Realizing the inefficiency of my initial methods, I sought to optimize the data collection process. I observed that storing 24 frames per second resulted in a significant amount of redundant data, as the color information did not change drastically on a per-second basis. Consequently, I reduced the frame extraction rate from 24 frames per second to 3 frames per second. This adjustment drastically cut down the number of frames without losing significant color data.

Additionally, instead of storing the frames themselves, I decided to extract the essential color information using the Color Thief library. Color Thief enabled me to identify the 10 most dominant colors from each frame and store these colors as RGB values. By saving only the RGB values as tuples, I significantly reduced the storage requirements.

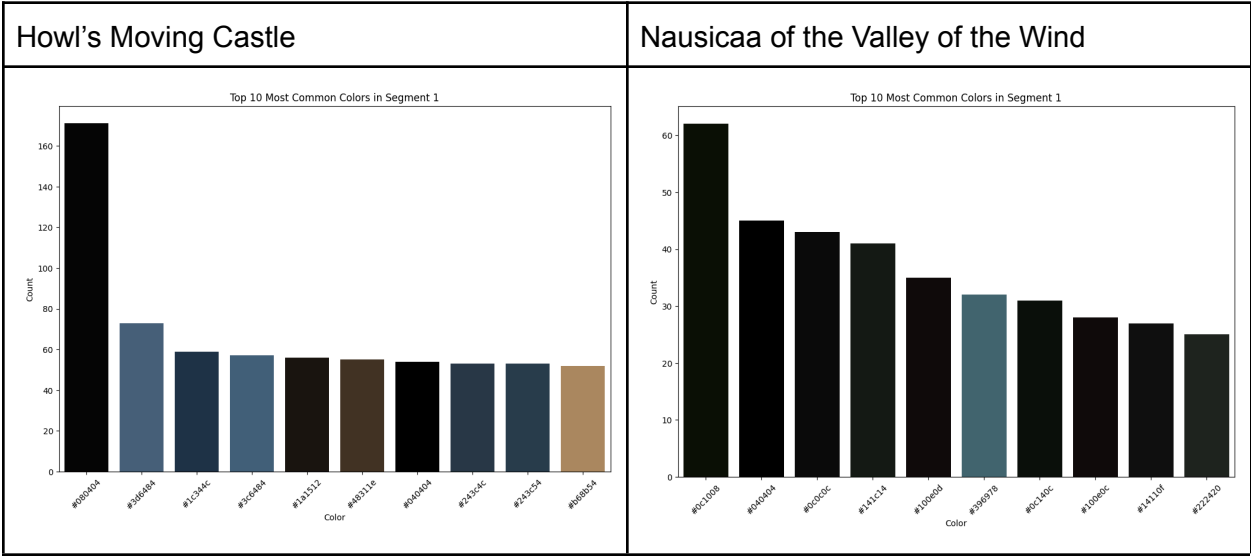
This optimized approach not only accelerated the data collection process but also ensured that I captured the necessary color data efficiently. The result was a CSV file containing the dominant RGB values from the film, which served as the foundation for further analysis.

Data Processing and Analysis

Overall Film Analysis

The first step in analyzing the collected data was to examine the 10 most dominant colors across the entire film. This involved aggregating the RGB values from all the frames and identifying the colors that appeared most frequently. By visualizing these dominant colors, I aimed to get an overarching view of the film’s color palette. This provided insights into the general color trends and themes present throughout the movie.

Top 10 dominant colors across the films (graph)



In Howls, there seems to be a mix of colors, but in Nuasicaa, there was a majority of dark tones only. I didn’t want to make any assumptions here yet, though, so I thought the next best thing was to segment the movie and see what the dominant colors were across different portions of the film.

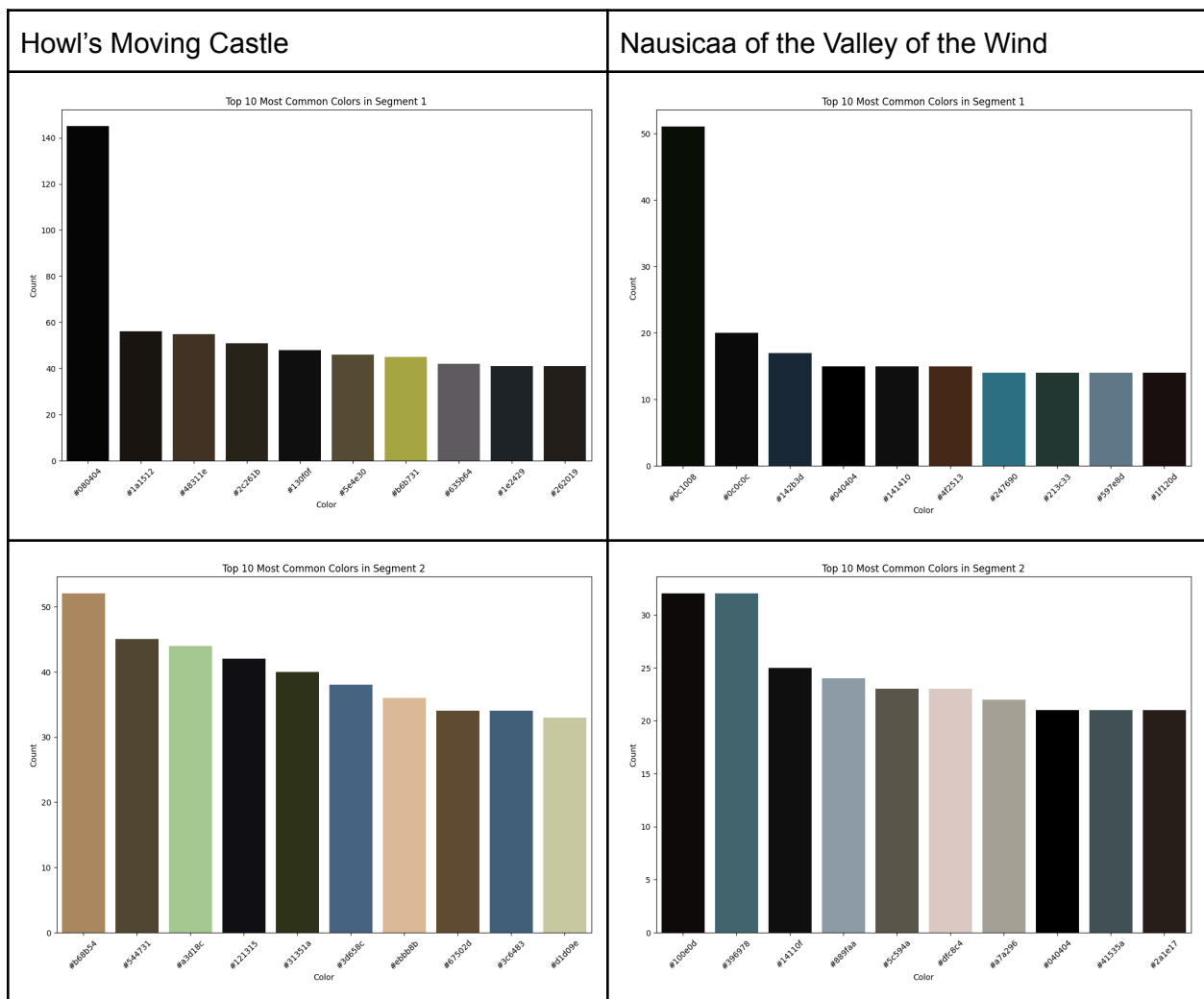
Segmented Analysis

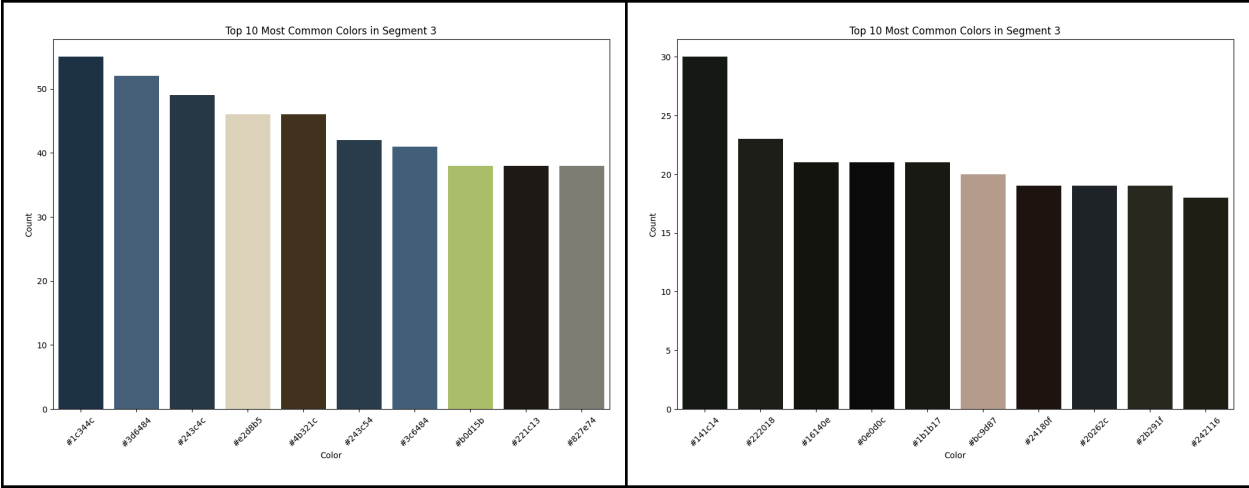
I performed this segmentation in two different ways:

Three Segments:

The film was divided into three equal parts, each representing a different portion of the story. By analyzing the dominant colors in each of these segments, I was able to observe how the color palette evolved as the narrative progressed.

Top 10 dominant colors across the films in 3 segments (graph)

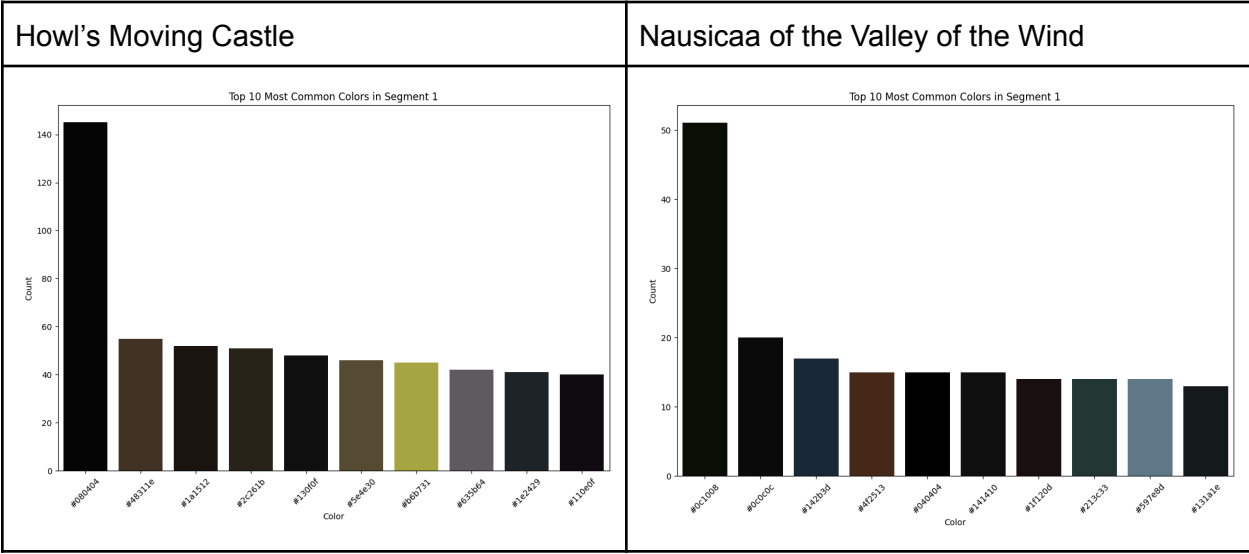


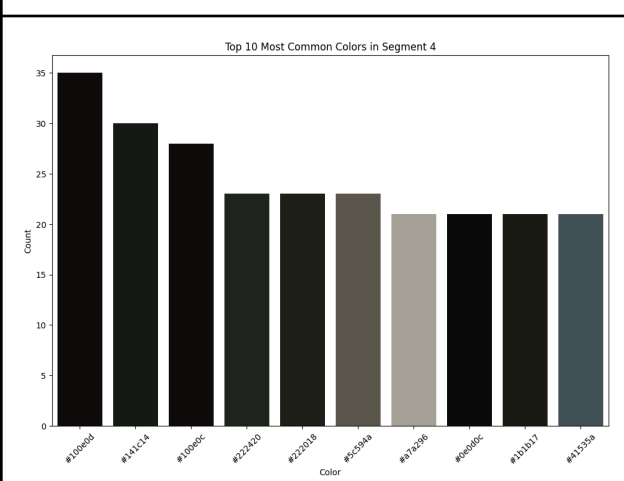
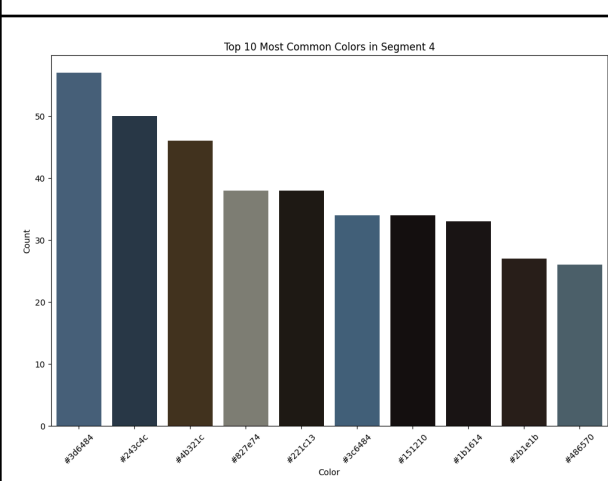
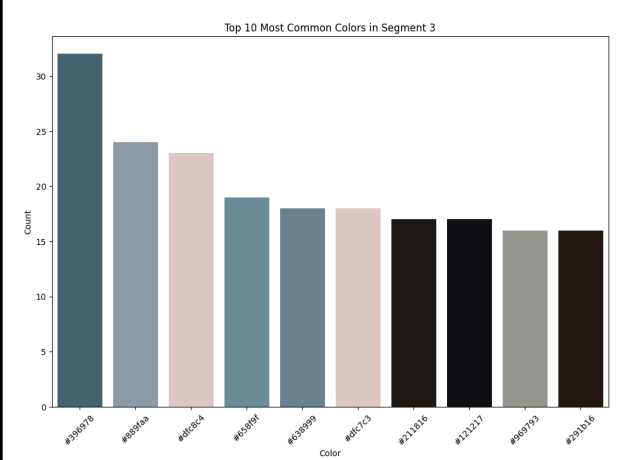
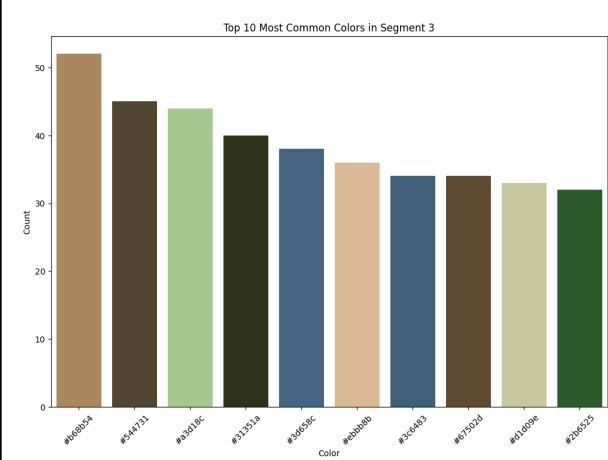
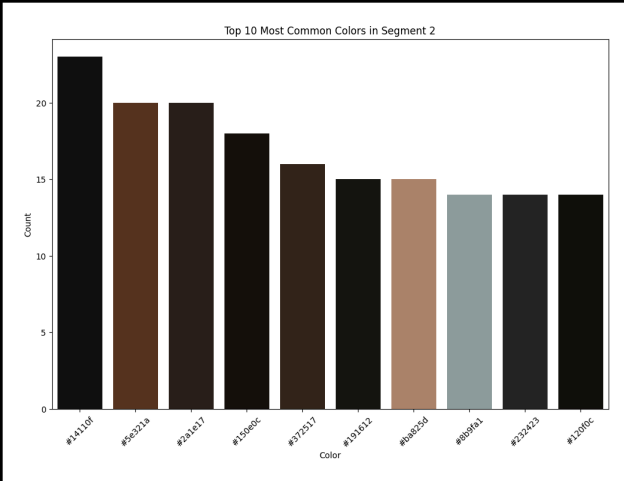
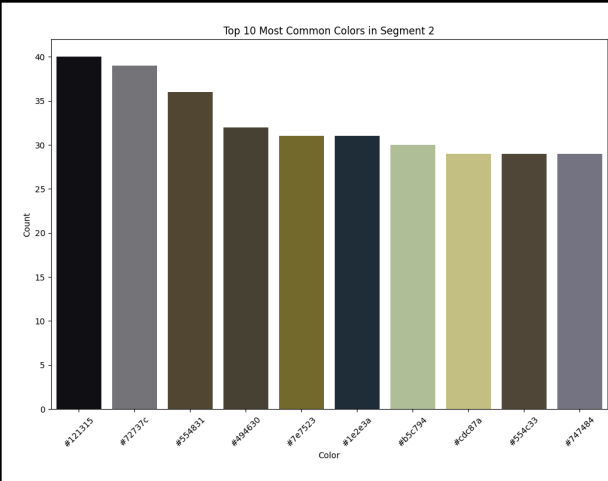


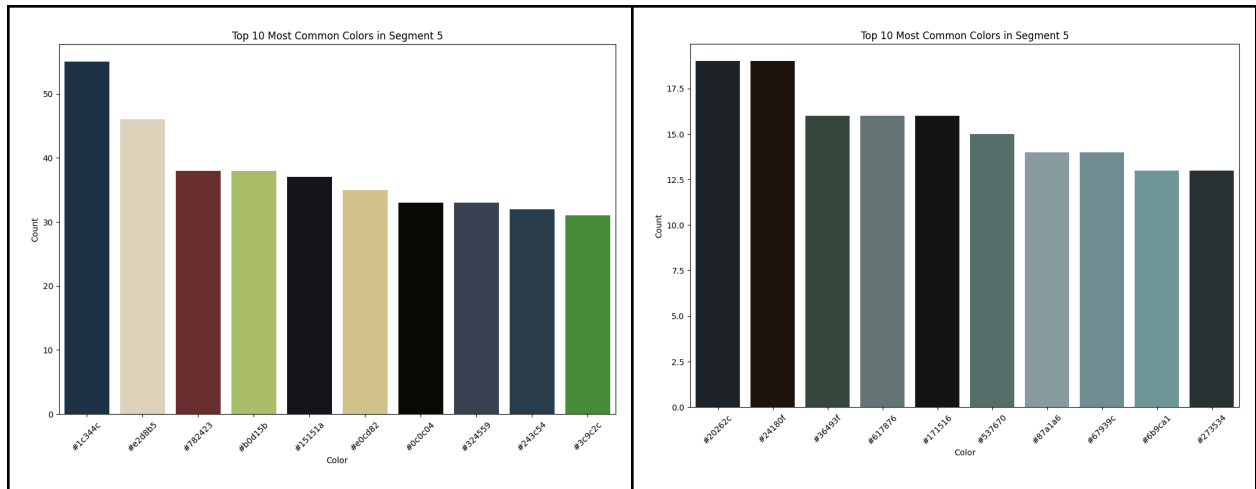
Five Segments:

In a more granular approach, the film was split into five segments. This allowed for a finer analysis of the color changes and helped identify any recurring color patterns in shorter intervals.

Top 10 dominant colors across the films in 5 segments (graph)





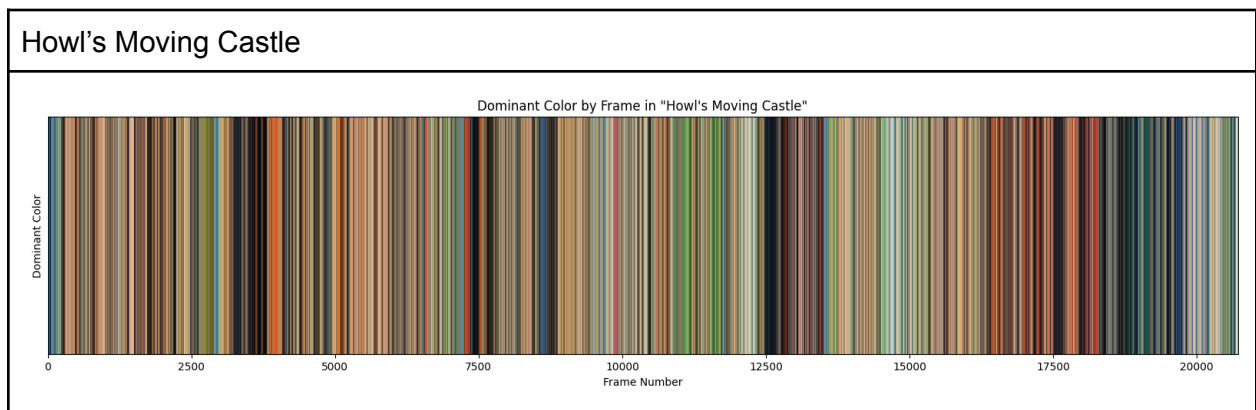


With these segmented analyses of the dominant colors, I was able to more clearly see the color palette evolve over the course of the film.

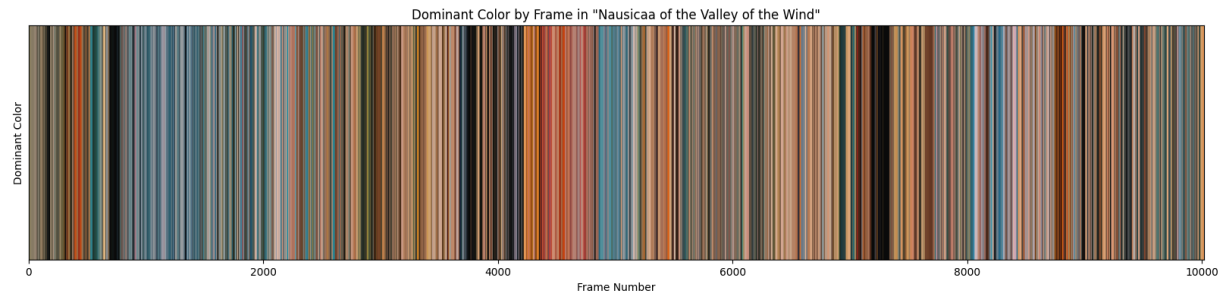
Frame-by-Frame Analysis

Now to capture the dynamic nature of the film's color palette, I conducted a frame-by-frame analysis. This involved plotting the most dominant color for each frame across the entire movie. By visualizing the dominant color per frame, I was able to see how the color palette shifted from one scene to the next. This analysis provided a detailed and continuous view of the color transitions throughout the film.

Plotting each frames dominant color (graph)



Nausicaa of the Valley of the Wind



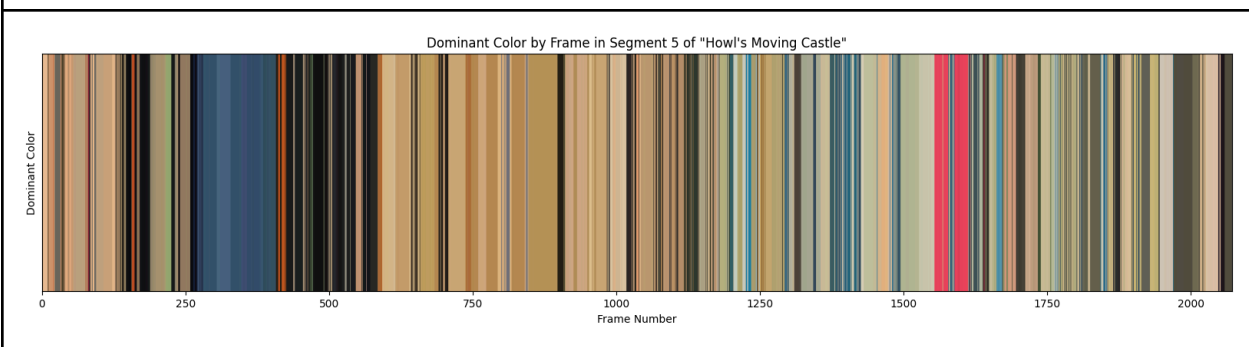
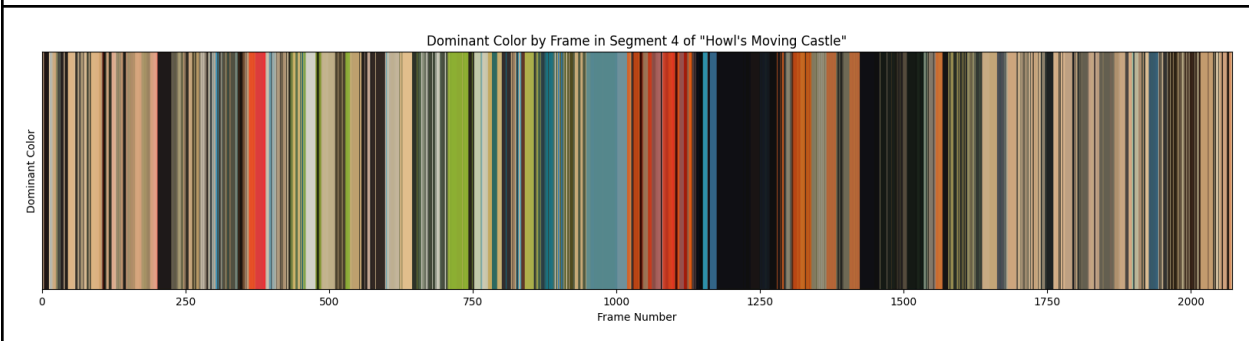
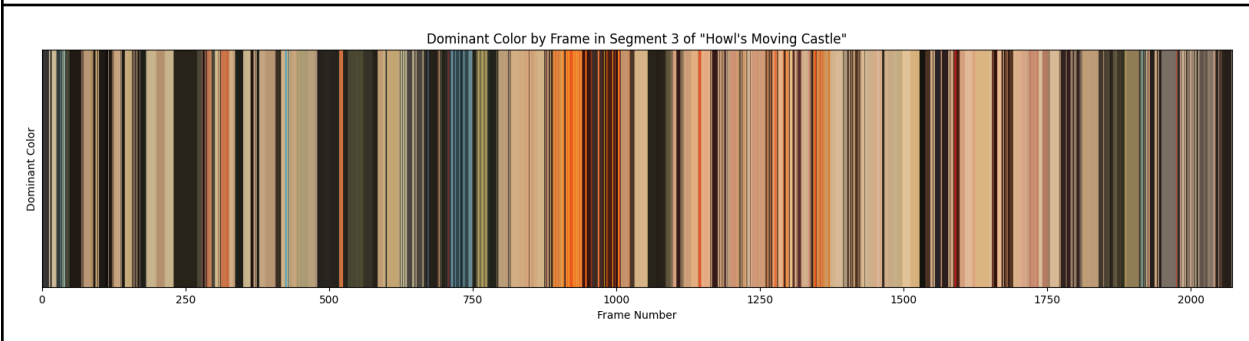
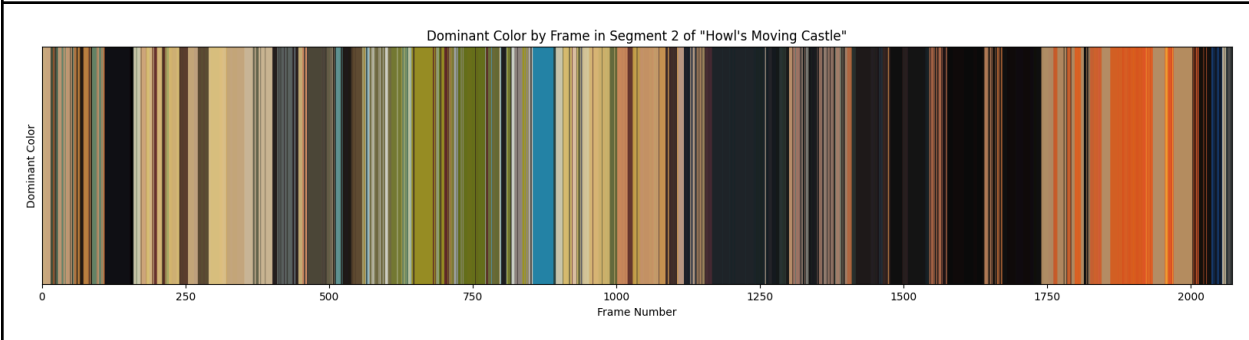
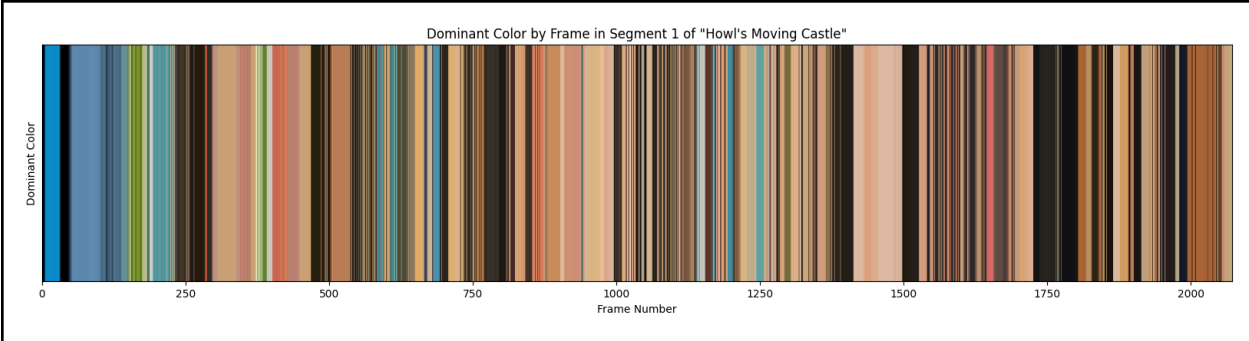
This was really cool, honestly. It clearly showed how the color evolved through the film's progression and was generally just a super interesting visual. From this, you can see that Howl's color palette emphasized different shades of green and red, whereas Nausicaa's emphasized reds/beiges and blues.

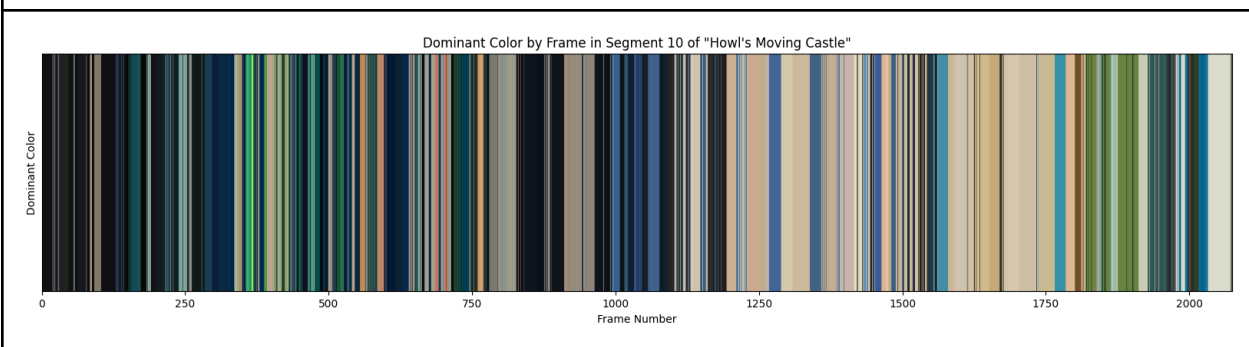
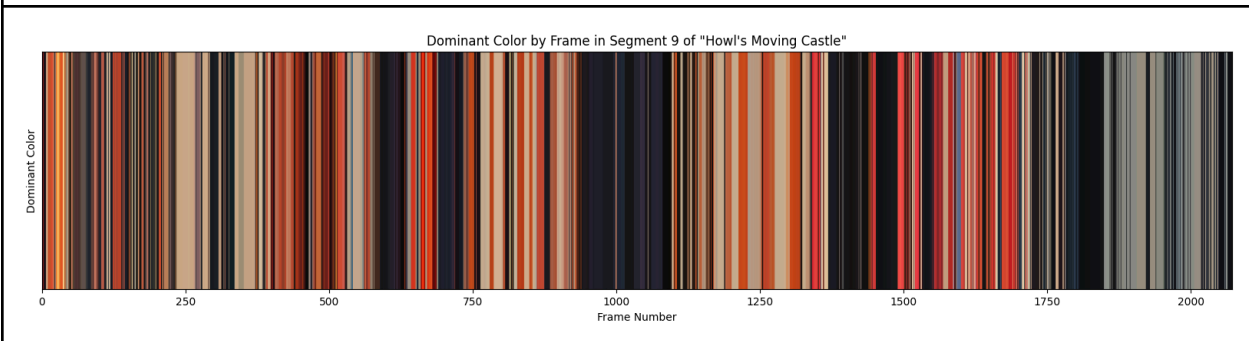
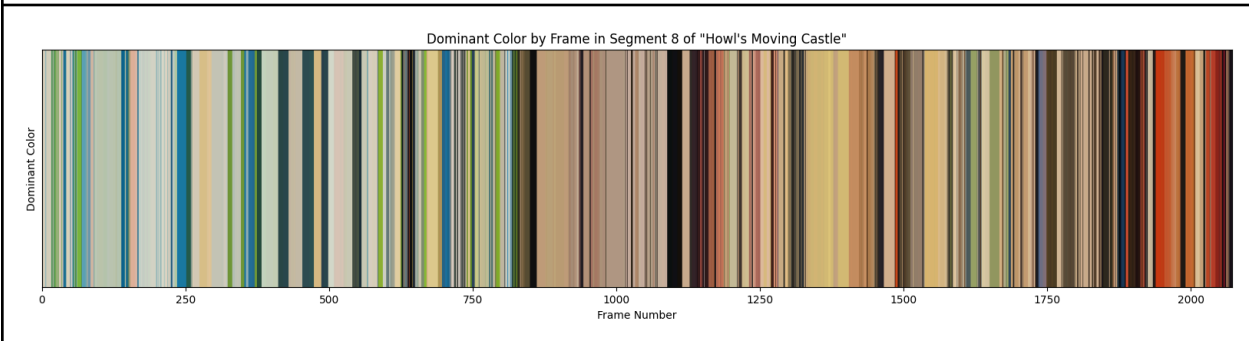
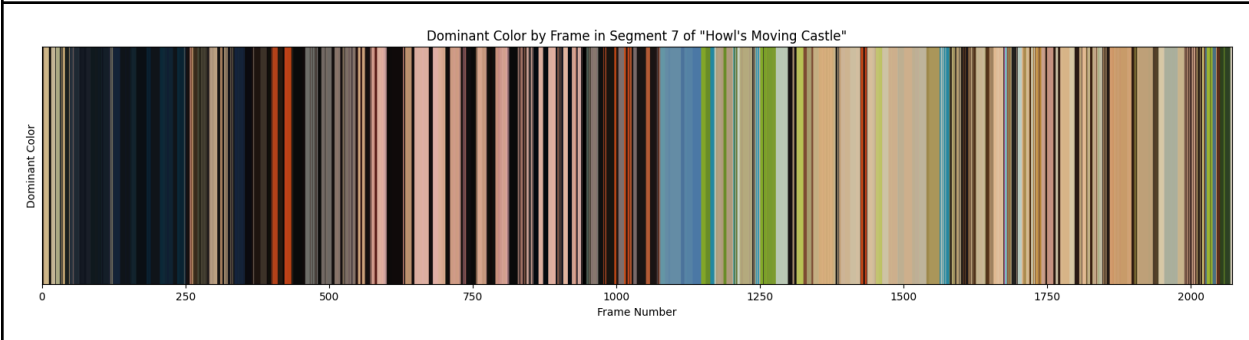
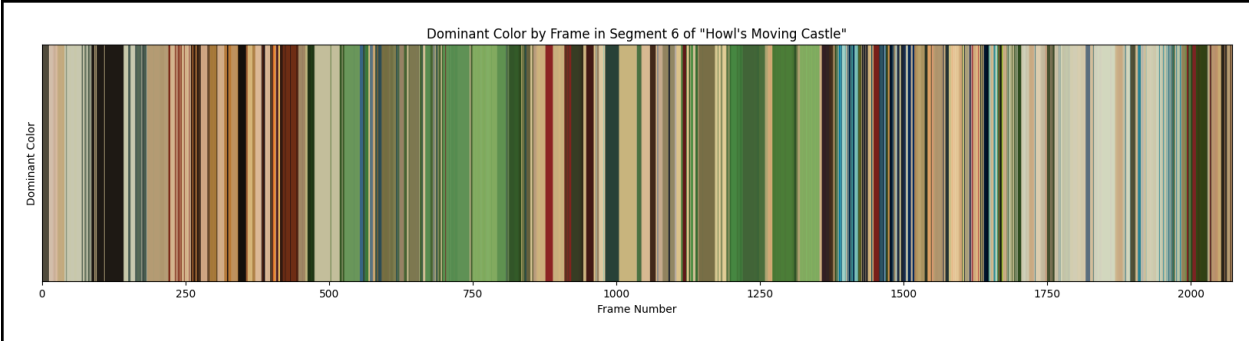
Segmented Frame Analysis

Building on the frame-by-frame analysis, I repeated this process with the film divided into ten segments. This segmented frame analysis aimed to understand the color trends in more specific portions of the film. By plotting the dominant colors for each frame within each segment, I could compare the color usage in different parts of the movie and identify any localized color patterns.

Howls - Plotting each frame's dominant color by segment (graph)

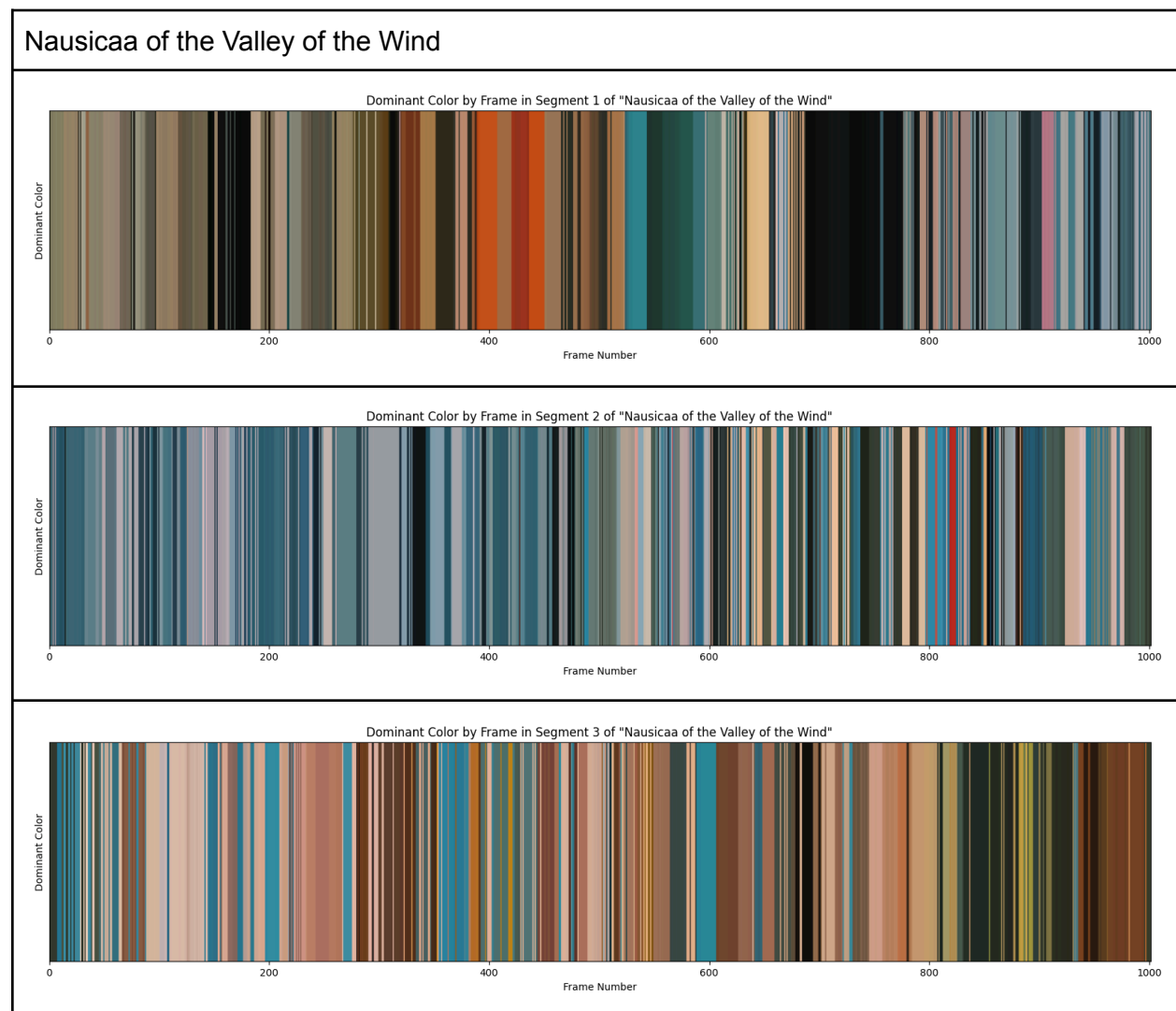
Howl's Moving Castle

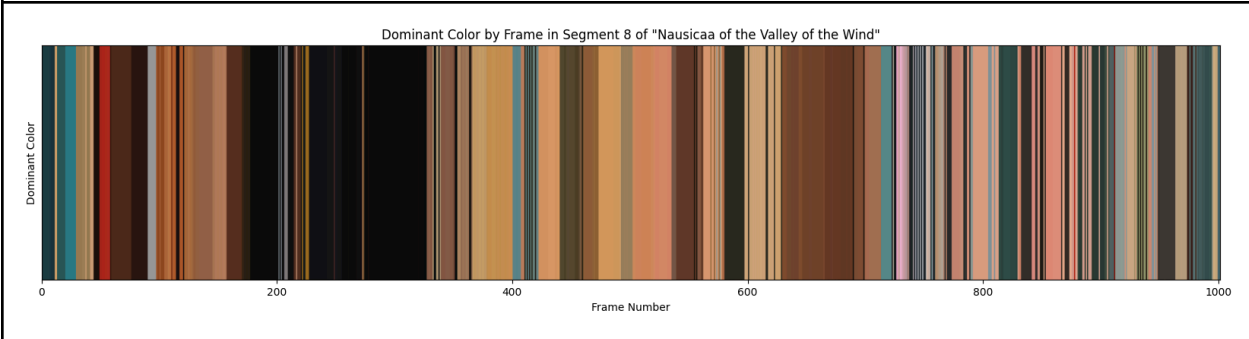
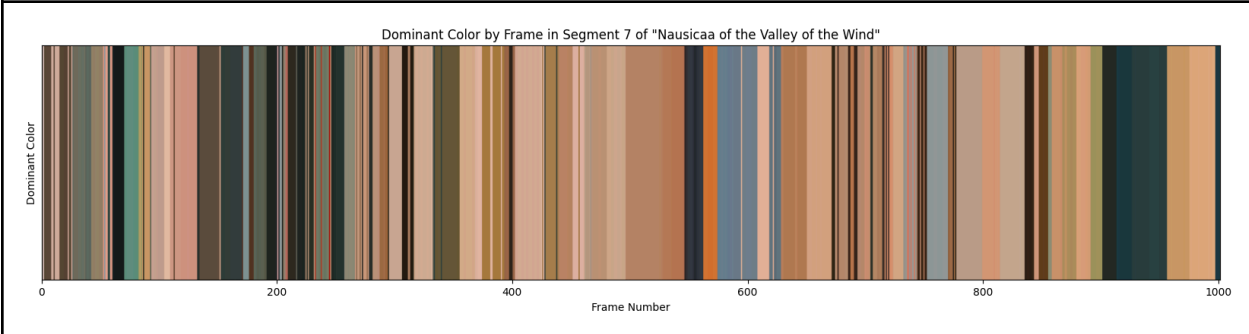
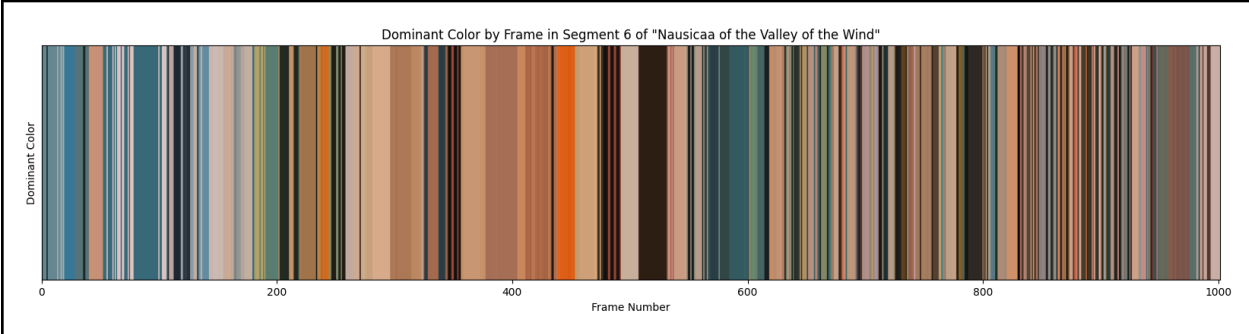
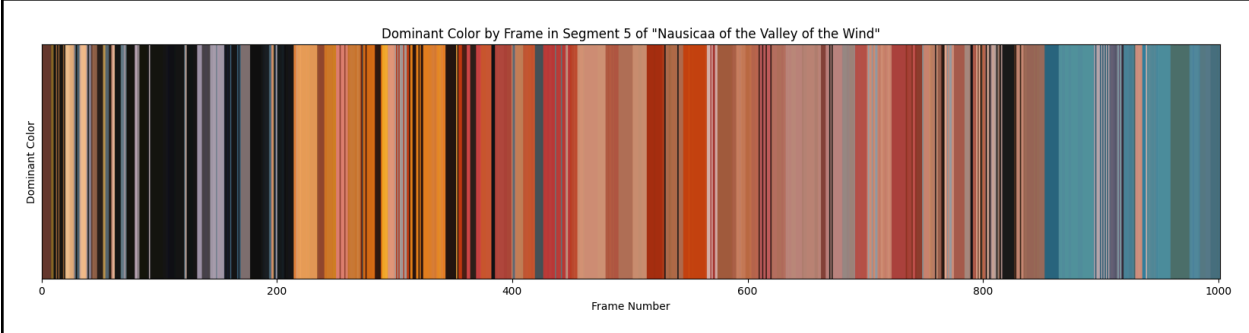
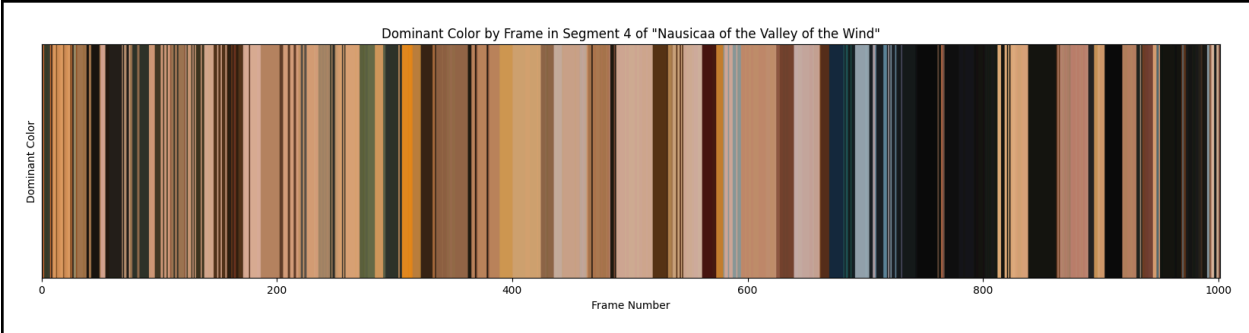


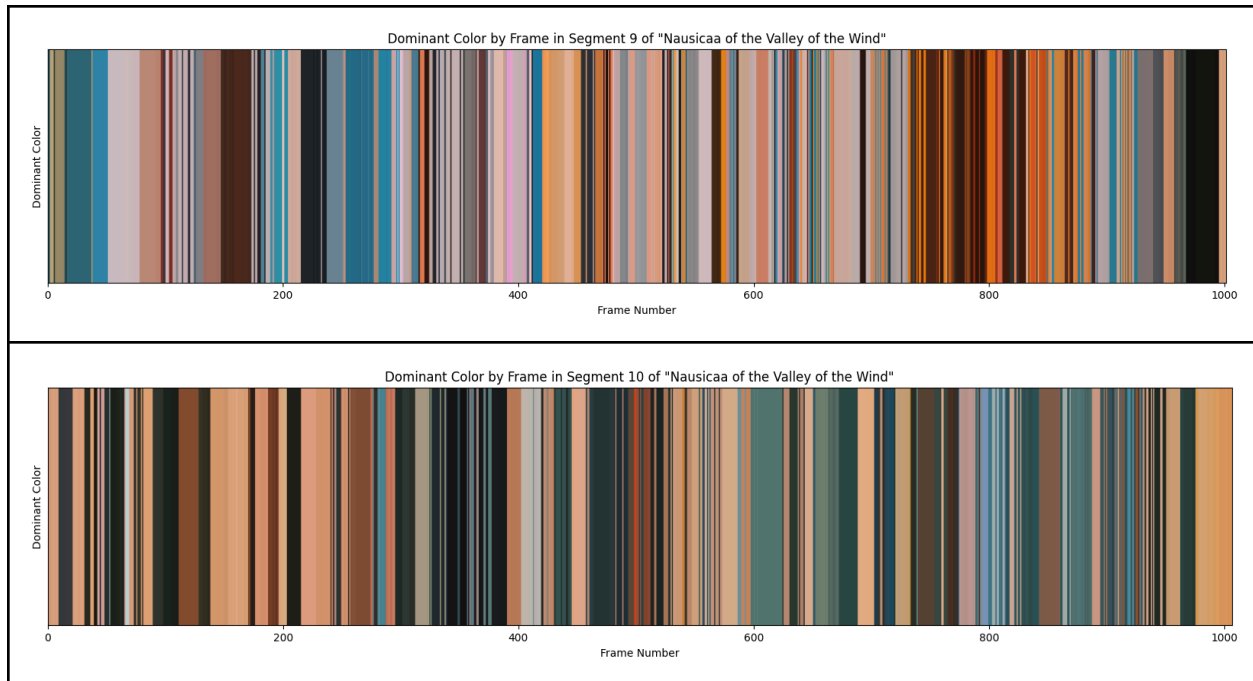


This was interesting, as breaking the film into segments allowed us to see how the colors matched the emotions evoked in each section. For example, the clearest segment is segment nine, when Howl joins the war and his castle/town is being bombed; you can see the anger/sorrow in the film segment matched by the deep hues of red and black. Similarly, around segment six, you can see bright shades of green and beige, likely signifying scenes in which they are by the garden or the vibrant colors of the capital.

Nausicaa - Plotting each frame's dominant color by segment (graph)







Again, this was super insightful as you could map sections of the movie and its emotions to the colors used in that section. You could see deep reds overtaking the color analysis when the Ohms were angry in segments one and three, and different shades of teal were prominent when they explored the overtaken forests, such as in segment two. Now that I could visually tell the color palettes encompassing each movie, I wanted to use a data science technique to give me a substantive answer as to which primary colors are found throughout the film.

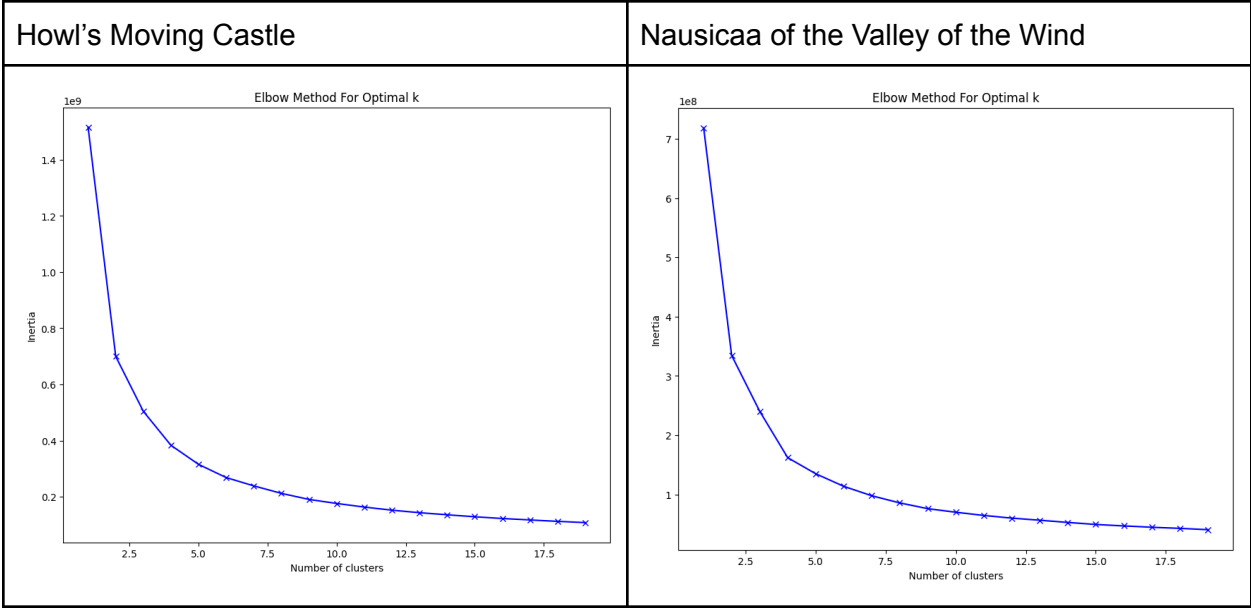
K-Means Clustering

So to produce a set of colors that represent the film, I decided to use K-Means clustering to determine the most dominant hues for each film.

Elbow Method:

Initially, I used the elbow method to determine the optimal number of clusters. The elbow method involves plotting the within-cluster sum of squares (inertia) against the number of

clusters and looking for an "elbow" in the plot. This elbow represents a point where adding more clusters does not significantly improve the compactness of the clusters. From my analysis, the elbow point suggested that 3 or 4 clusters would be optimal for both Howl's and Nausicaa, as this was where the inertia started to level off.

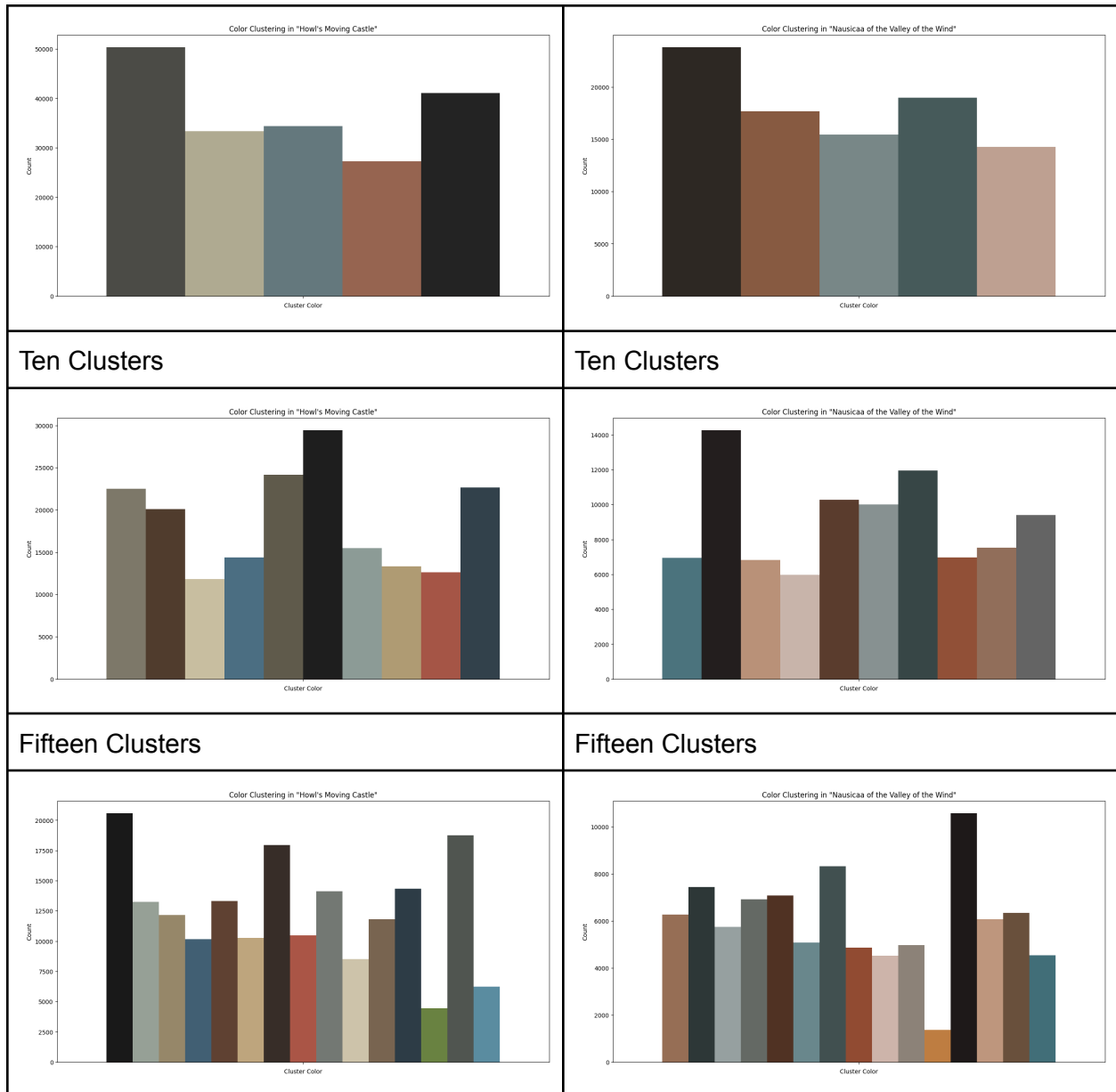


Cluster Analysis:

However, I observed that fewer clusters tended to produce muted colors, which defeats the purpose of capturing the film's vibrant palette. To confirm this, I checked the clustering output at five, ten, and fifteen clusters to see what the colors looked like:

Clustering output at 5,10,15 clusters (graph)

Howl's Moving Castle	Nausicaa of the Valley of the Wind
Five Clusters	Five Clusters

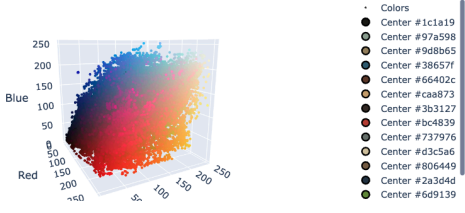
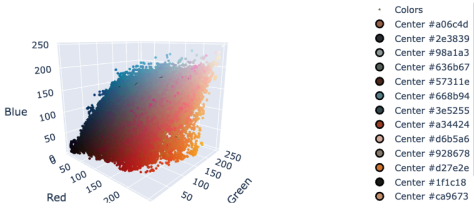
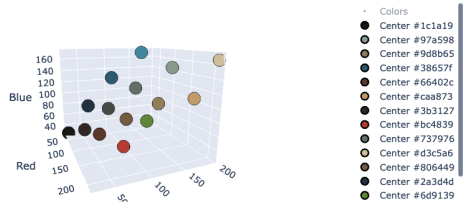
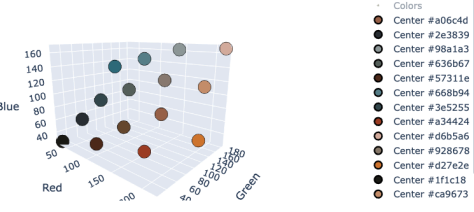


Increasing the number of clusters allowed for a more detailed representation of the film's color palette. At 5 clusters, the colors were somewhat generalized, but as I increased to 10 and 15 clusters, the diversity and richness of the colors became more apparent. This confirmed that a higher number of clusters provided a better depiction of the nuanced color variations present in the film.

3D RGB Space Visualization:

To visualize the clusters, I plotted the 15 clusters in 3D RGB space. Each cluster center was represented as a point in the RGB color space, allowing me to see how the colors were distributed and grouped. This visualization helped in understanding the relationships between different colors and how they were clustered together. I first performed this visualization on the entire film, and then the film segmented into ten parts, providing a comprehensive view of the color distribution throughout the movie.

Entire Film K-Means Clustering (graph)

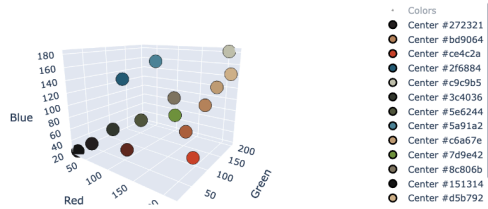
Howl's Moving Castle	Nausicaa of the Valley of the Wind
All Colors Plotted	All Colors Plotted
Color Clustering in "Howl's Moving Castle" 	Color Clustering in "Nausicaa of the Valley of the Wind" 
Just the Cluster Centers	Just the Cluster Centers
Color Clustering in "Howl's Moving Castle" 	Color Clustering in "Nausicaa of the Valley of the Wind" 

Segmented Film K-Means Clustering (graph)

Howl's Moving Castle	Nausicaa of the Valley of the Wind
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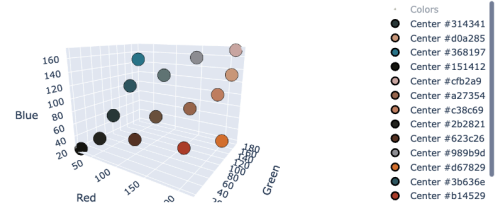
Just the Cluster Centers

Color Clustering in Segment 1 of "Howl's Moving Castle"

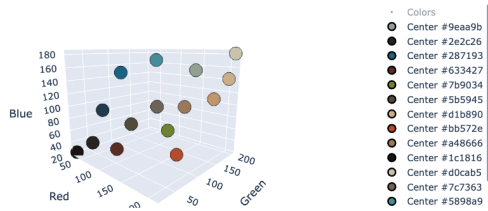


Just the Cluster Centers

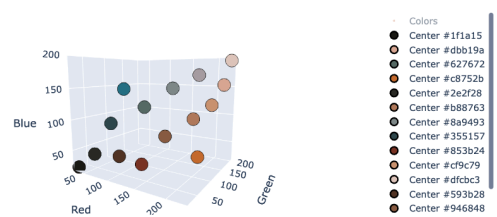
Color Clustering in Segment 1 of "Nausicaa of the Valley of the Wind"



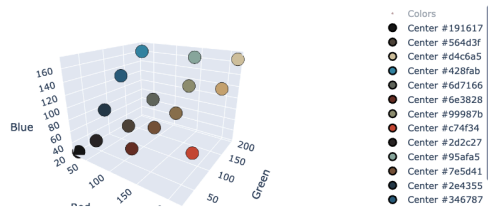
Color Clustering in Segment 2 of "Howl's Moving Castle"



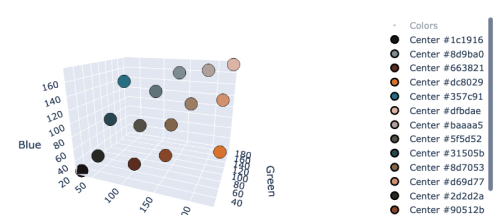
Color Clustering in Segment 2 of "Nausicaa of the Valley of the Wind"



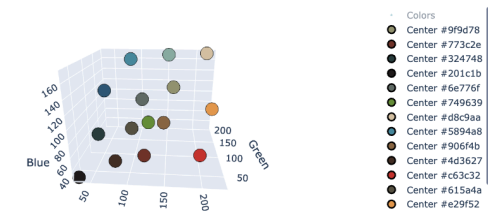
Color Clustering in Segment 3 of "Howl's Moving Castle"



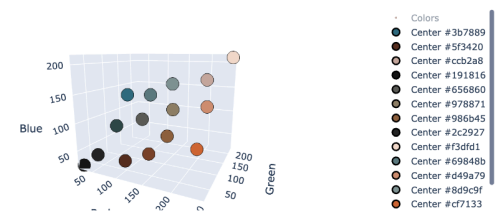
Color Clustering in Segment 3 of "Nausicaa of the Valley of the Wind"



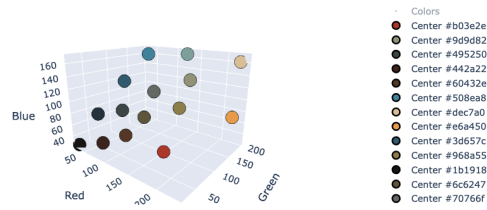
Color Clustering in Segment 4 of "Howl's Moving Castle"



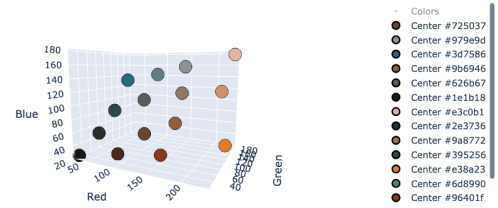
Color Clustering in Segment 4 of "Nausicaa of the Valley of the Wind"



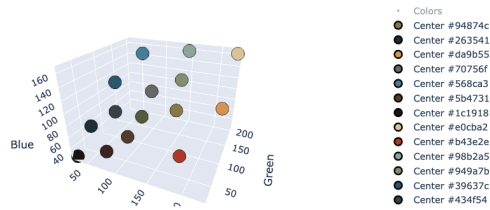
Color Clustering in Segment 5 of "Howl's Moving Castle"



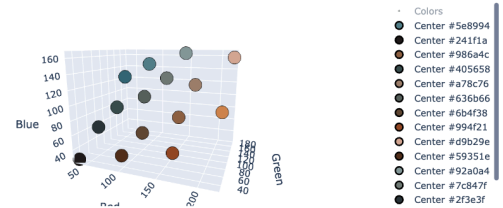
Color Clustering in Segment 5 of "Nausicaa of the Valley of the Wind"



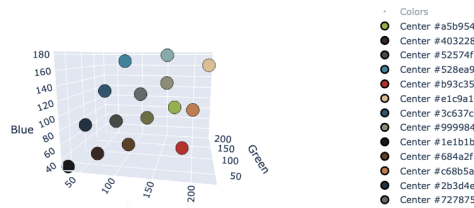
Color Clustering in Segment 6 of "Howl's Moving Castle"



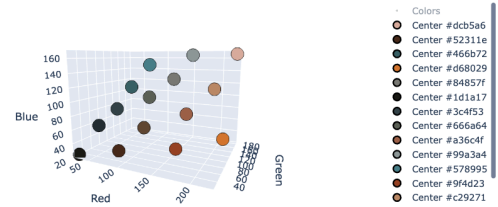
Color Clustering in Segment 6 of "Nausicaa of the Valley of the Wind"



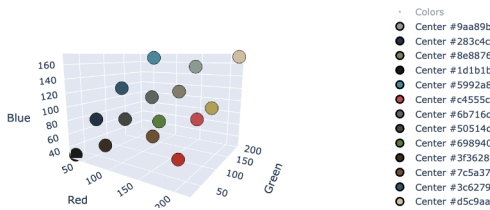
Color Clustering in Segment 7 of "Howl's Moving Castle"



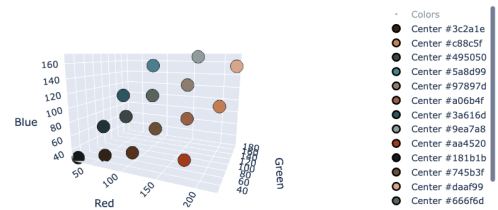
Color Clustering in Segment 7 of "Nausicaa of the Valley of the Wind"

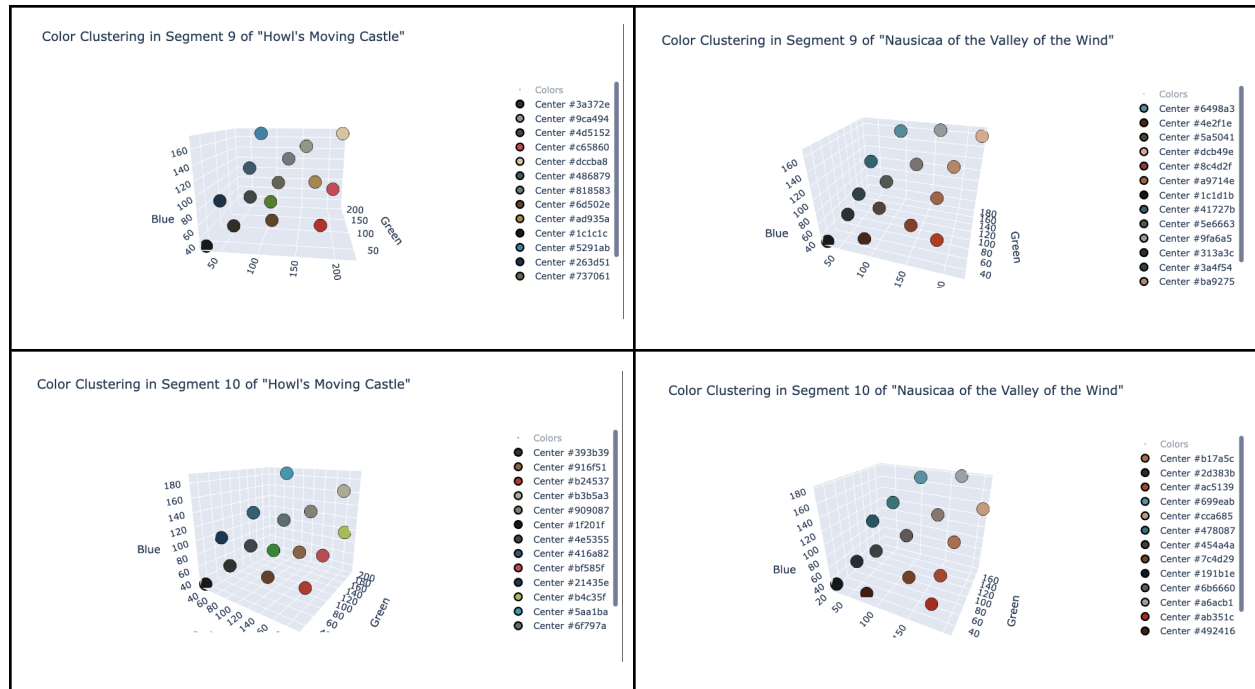


Color Clustering in Segment 8 of "Howl's Moving Castle"



Color Clustering in Segment 8 of "Nausicaa of the Valley of the Wind"



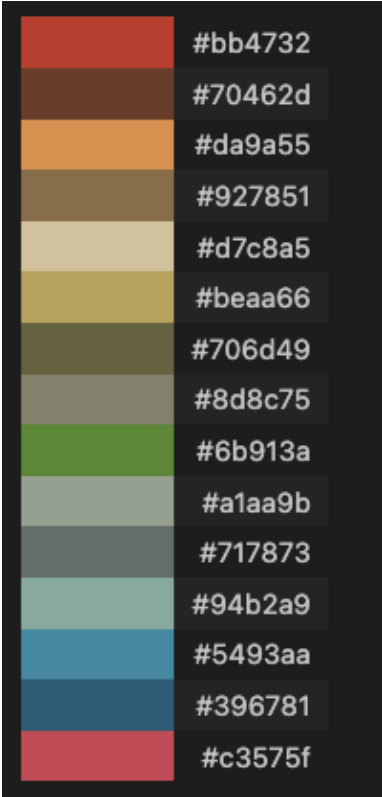
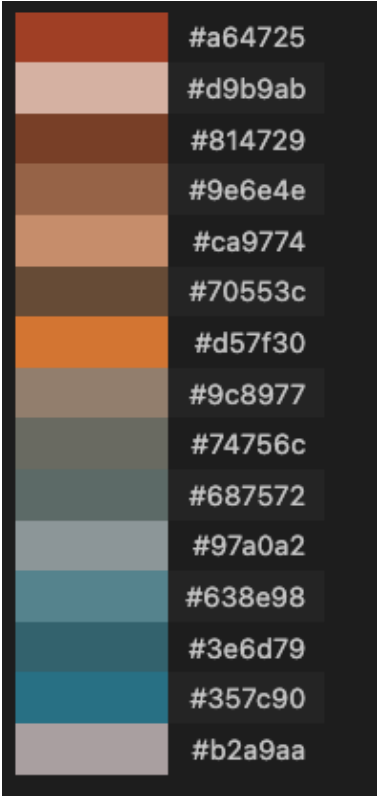


Refined Clustering:

To refine the color representation, I re-ran k-means on the 150 cluster centers derived from the 10 segments, aiming to get 15 representative colors. This step involved clustering the pre-identified cluster centers to further condense the color data. Additionally, I applied a brightness threshold to exclude overly dark colors, ensuring a balanced and aesthetically pleasing color palette. This final step was crucial in achieving a representative set of colors that accurately reflected the film's vibrant and varied hues.

Final Color Palette

The result of these analyses was a set of 15 representative colors that encapsulate the essence of each film's color palette. These colors serve as the foundation for creating fashion outfits inspired by the movie, reflecting the vibrant and diverse color themes present throughout the film.

Howl's Moving Castle	Nausicaa of the Valley of the Wind
 #bb4732 #70462d #da9a55 #927851 #d7c8a5 #beaa66 #706d49 #8d8c75 #6b913a #a1aa9b #717873 #94b2a9 #5493aa #396781 #c3575f	 #a64725 #d9b9ab #814729 #9e6e4e #ca9774 #70553c #d57f30 #9c8977 #74756c #687572 #97a0a2 #638e98 #3e6d79 #357c90 #b2a9aa

Created Outfits

All the outfits showcased below were created by me, inspired by the 15-color clusters identified through the analysis. My goal was to design outfits that utilized these specific colors, staying as true to the determined shades as possible. However, due to the limitations of my wardrobe, some outfits may feature colors that are a few shades off from the original clusters. In cases where I lacked matching lighter tones, I used white or light beige as base colors.

Overall, this project was incredibly enjoyable, allowing me to blend my creative skills with data-driven insights to craft unique outfits. Designing outfits for the "Nausicaä of the Valley of the Wind" color scheme posed a greater challenge due to the more unique and similar tones,

making it difficult to work within my wardrobe constraints. Despite these challenges, I believe I created interesting and visually appealing outfits. I aim to create more outfits when I gain access to my winter wardrobe, so for now, please enjoy my summer wardrobe attempt.

Howl's Created Outfits (Images)



Howl's Moving Castle
COLORS USED

#5493AA

#6B913A

#D7C8A5

To be fair, I used a lighter shade as the t-shirt, but the Uniqlo U Airism beige T would've been perfect here.

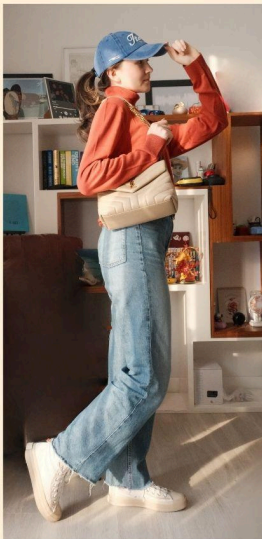




Howl's Moving Castle
COLORS USED

#396781

#C3575F



Howl's Moving Castle
COLORS USED

#BB4732

#5493AA

#D7C8A5



Howl's Moving Castle COLORS USED

#DA9A55

#BEAA66

#706D49

#396781

#5493AA

#70462D

A slightly lighter shade is actually used in the shirt.



Howl's Moving Castle COLORS USED

#94B2A9

#C3575F

#FFFFFF

I didn't have this exact color, so I went with a slightly lighter wash of blue. It still fits into the general color scheme of the movie, even if not a cluster center color.



Howl's Moving Castle COLORS USED

#D7C8A5

#D7C8A5

↗
The pants use slightly lighter beige, which is still well within the confines of the film's colors.



Howl's Moving Castle COLORS USED

#FFFFFF

#396781

Outfit Reference



Nausicaa's Created Outfits (Images)



Nausicaa of the Valley
of the Wind
COLORS USED

#A64725

#D9B9AB

#638E98



Nausicaa of the Valley
of the Wind
COLORS USED

#CA9774

#357C90



Nausicaa of the Valley of the Wind COLORS USED

#357C90

#D9B9AB

#814729

Technically, the pants are a bit lighter of a shade but still within the confines of the movie - e.g., the sands underneath the forest of decay.



Nausicaa of the Valley of the Wind COLORS USED

#D57F30

#638E98



Nausicaa of the Valley
of the Wind
COLORS USED

#A64725

#9E6E4E

#FFFFFF

Not specifically in the clusters, but it works as the
sands underneath the forest of decay.